

Towards Bridging Video and Language by Caption Generation and Sentence Localization

Shaoxiang Chen

sxchen13@fudan.edu.cn

Shanghai Key Lab of Intelligent Information Processing,
School of Computer Science, Fudan University
Shanghai, Shanghai, China

ABSTRACT

Various video understanding tasks (classification, tracking, action detection, etc.) have been extensively studied in the multimedia and computer vision communities over the recent years. While these tasks are important, we think that bridging video and language is a more natural and intuitive way to interact with videos. Caption generation and sentence localization are two representative tasks for connecting video and language, and my research is focused on these two tasks. In this extended abstract, I present approaches for tackling each of these tasks by exploiting fine-grained information in videos, together with ideas about how these two tasks can be connected. So far, my work have demonstrated that these two tasks share a common foundation, and by connecting them to form a cycle, video and language can be more closely bridged. Finally, several challenges and future directions will be discussed.

CCS CONCEPTS

• Computing methodologies → Computer vision.

KEYWORDS

video caption generation; temporal sentence localization

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1 INTRODUCTION

Video caption generation [31] (or video captioning) aims at generating a sentence to describe a video's content, and sentence localization in videos [11] is to temporally localize a video segment which matches with a given sentence description. Together these two tasks connect video and language from both directions. As for application, video captioning can be the underlying technique for automatic video commenting and assistance for the visually impaired, and sentence localization can empower a more useful text-based video search engine.

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The recent success of deep learning in computer vision and natural language processing have driven the advancements of video captioning and temporal sentence localization in videos. The models in most deep learning-based video captioning methods [6, 25, 29, 31, 33, 43] are organized in an encoder-decoder style. The encoder is responsible for extracting and aggregating information from videos, and is the most important part of a captioning model. A large portion of existing methods only use coarse-grained visual representations in the encoder, which limits their ability to understand the video contents. This is also true for temporal sentence localization. Parts of my research [6, 8] aim at exploiting fine-grained information in videos to obtain better video representations for the video-and-language tasks.

Recently, I try to jointly study caption generation and sentence localization. Since a video contains multiple modalities, one of the common foundations of these tasks is utilizing multimodal features. Multimodal fusion is actually a popular strategy used in video captioning [13, 19, 32, 37, 41], but existing methods use simple feature concatenation or modality attention, so the interactions between modalities are not enough for information communication. One of my works [5] designs a novel modality interaction method to better exploit the cross-modal complementary information. On the other side, I also try to bridge the caption generation module and the sentence localization module. This is necessary to solve the recently proposed weakly supervised dense event captioning task. We aim to introduce more information passing pathways between the two modules to enhance both of them and improve the final dense captioning performance.

2 RELATED WORK

Video Captioning. Video captioning methods mainly focus on obtaining better video representations, this can be done by exploiting temporal relations, multimodal features, and spatial structures. Early methods treat video captioning as a sequence-to-sequence translation problem [31], and exploiting the temporal structure [28, 29, 43] within a video helps more efficiently utilizing the video content. Utilizing the complementary information among multimodal features is a popular strategy adopted by most video captioning methods. However, the multimodal fusion mechanism is still simple. [1, 4, 26, 37] use feature concatenation along the channel dimension, and [12, 13, 19, 41, 51] use attention-based feature fusion. Spatial structures can be modeled by spatial attention or object relations. Spatial attention-based method [6, 15, 34, 42] aim at discovering the important regions of interest, and by focusing more on these regions, they can better understand the video content. More recent methods [24, 49, 53, 54] have chosen to start with explicit object regions and further model the object relations.

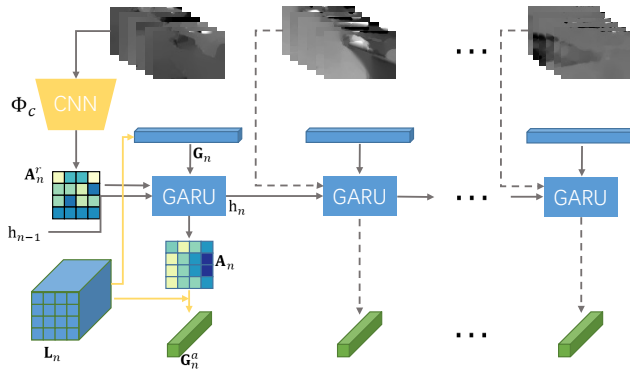


Figure 1: The Motion Guided Spatial Attention (MGSA) framework for video captioning. The spatial information is aggregated under the guidance of motion information (stacked optical flow). The readers can refer to [6] for the complete framework.

Temporal Sentence Localization. Various proposal generation strategies have been adopted by temporal sentence localization methods. The simplest strategy is sliding window [7, 11, 18, 38, 52], which does not require any learnable parameters. Anchor-based proposal generation [3, 36, 39, 44, 46, 47] means generating multi-scale proposals at each sampled time point. It requires a learnable module and predefined anchor scales. Recently, a type of anchor-free methods [17, 23, 50, 55] model the starting and ending point, and combine them to predict the probabilities of every possible temporal segment. There are also some methods [8, 20, 21, 45, 48] that only produce one prediction by regression, and their advantage is a simpler training process and faster convergence.

3 PROPOSED APPROACHES

My works are mainly focused on exploiting fine-grained information in videos and further bridging the captioning and localization tasks. In this section, we will introduce two methods for exploiting fine-grained spatial information for video captioning and sentence localization, respectively. We will also introduce how to bridge the two tasks via a better modality interaction method and by passing information between captioner and localizer.

3.1 Exploiting Fine-Grained Information

3.1.1 Motion Guided Spatial Attention for Video Captioning. For video captioning, utilizing spatial information within frames is crucial to understanding video contents and generating better captions. Most methods adopt a pre-trained 2D CNN to extract spatial information at its global average pooling layer, thus no spatial structure is preserved, i.e., all the spatial locations are deemed equally important. This is counterintuitive, since the key objects and their movements are more informative than the background regions.

To avoid this problem, we proposed Motion Guided Spatial Attention (MGSA), which is an adaptive spatial attention mechanism. As shown in Figure 1, for each frame, the spatial feature map L_n is extracted by a pre-trained 2D CNN and later aggregated by a spatial attention map A_n . One contribution of this work is using motion as guidance for the spatial attention. The intuition behind this is

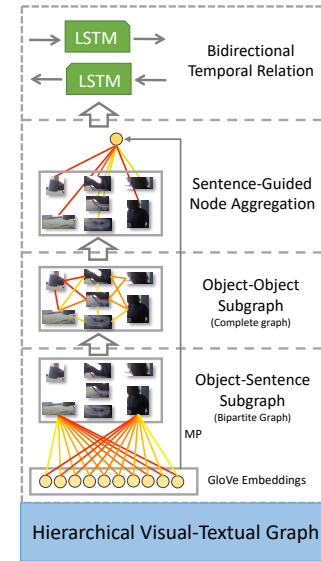


Figure 2: The Hierarchical Visual-Textual Graph framework used in our sentence localization method [8]. The readers can refer to [8] for the complete framework.

that regions with apparent motion are usually (but not always) more related to the event in the video. We chose optical flow to capture short-term motion around the sampled key frames, and apply a custom CNN to the stacked optical flow images to transform them into an attention map. Besides, we devised a Gated Recurrent Attention Unit to establish temporal relations between the spatial attention maps. The reason is that the adjacent attentions are not dramatically different, and incorporating the previous attention map into the current one helps stabilize attention maps and leads to better captioning performance.

3.1.2 Hierarchical Visual-Textual Graph for Sentence Localization. Fine-grained spatial information is also important for solving temporal sentence localization, since the sentence description is complex and we need to model the interactions between objects in order to match the sentence to the video frames. However, we can not rely on external object detectors to provide the object semantics (or classes). Because they are pre-trained on datasets with a limited number of classes and can not transfer sufficient semantic information for this task.

To address this problem, we only use the object regions from the pre-trained object detector and construct a Hierarchical Visual-Textual Graph to incorporate rich semantic information and model object interaction. As shown in Figure 2, each object in a frame first interacts with the word embeddings to absorb semantic information related to itself. The objects with correct semantics then interact with each other in a fully connected graph, and each edge weight in the graph is computed from the two connecting objects to decide how much they are related. The set of interacted objects are then reduced into a single feature vector by their relevance to the entire sentence query, so that the frame representation is more compact. Finally, the temporal relation across frames is established by a bidirectional LSTM. This is the feature encoding part of the whole framework, the readers can refer to [8] for the complete method.

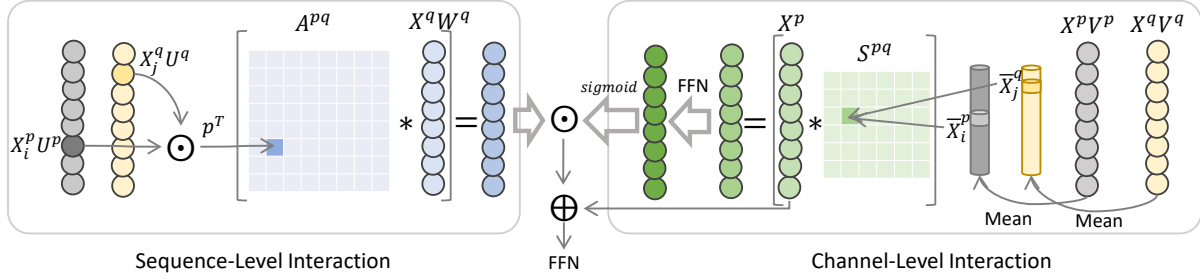


Figure 3: The sequence-level and channel-level modality interaction of our proposed method [5]. The interaction is performed for each pair of modalities from a video. The readers can refer to [5] for the complete framework.

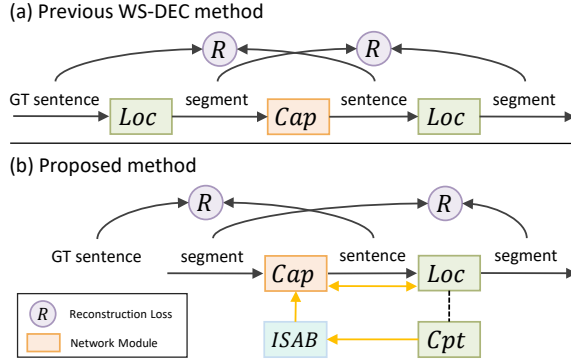


Figure 4: The comparison of previous WS-DEC method (a) and our proposed method (b). Ours has more pathways for information passing between the captioner and the localizer. The readers can refer to [9] for the complete framework.

3.2 Bridging The Two Tasks

Video captioning and temporal sentence localization can share the same video feature encoder, and their models can be connected to form a cycle that translates video to language and back. In this section, we introduce two methods that bridge the two tasks.

3.2.1 Learning Modality Interaction. How to better utilize the multiple modalities in videos is a popular research topic in video-related tasks. For video captioning, previous works have proved fusing multimodal features can consistently and effectively improve performances. However, they commonly use feature concatenation or modality-wise attention to fuse multimodal features, and the problem is that the features do not interact (no information communication across modality) before fusion. So the complementary information from other modalities are not fully exploited.

Our proposed method [5] aims at learning a better modality interaction for utilizing the complementary information between each pair of modalities. As shown in Figure 3, for each modality pair (e.g., visual and audio), we model the relations between each pair of feature vectors from the two feature sequences. This dense interaction ensures that each feature vector in one modality has a complete receptive field when absorbing information from another modality. In addition, we model the channel-level interaction between the mean-pooled features of both modalities and use the result to construct a channel gate to control the feature channels

Model	B@4	C	M
Spatial Attention	49.8	72.2	32.9
Spatial Attention w/ GARU	51.0	81.8	34.0
MGSA w/o GARU	51.0	83.3	33.1
MGSA w/ GARU	53.0	86.4	34.7

Table 1: Comparison of MGSA variants on the MSVD dataset.

#	OSS	OOS	SGNA	IoU=0.3	IoU=0.5	IoU=0.7
0	✗	✗	✗	55.56	40.38	19.49
1	✓	✗	✗	58.36	44.97	21.77
2	✗	✓	✗	57.42	42.31	19.97
3	✗	✗	✓	58.46	42.93	20.86
4	✓	✓	✓	61.37	47.27	23.30

Table 2: Performances of our model variants with different HVTG configurations. OSS means object-sentence subgraph, OOS means object-object subgraph, and SGNA means sentence-guided node aggregation.

of the sequence interaction results. The interaction results of all the possible modality pairs are finally fused by a position-wise learnable attention. As we demonstrated in the paper, the resulting feature can be used for downstream tasks like video captioning and sentence localization.

3.2.2 Bridging Captioner and Localizer. A video captioning model translates a video into a sentence, and a sentence localization model can return a video segment given a query sentence. Together they can form a cycle that starts with video, goes to sentence, and back to video. And by applying cycle consistency constraint, this procedure can be used to solve the weakly supervised dense event captioning (WS-DEC) task [10]. But current WS-DEC methods [10] only rely on directly connecting the captioner and localizer for information passing. This can only pass limited information (and gradient) between the two modules and can hurt the final performance.

In order to pass richer information between the captioner and localizer and improve the final captioning quality, we introduce several other pathways for information passing. As shown in Figure 4, we use a concept learning module as the basis for the localizer, and the latent concept features learned by this module are passed to an improved Induced Set Attention Block (ISAB). The ISAB refines the concept features and passes them to the captioner to help construct better video features. In addition, the captioner will produce sentence confidence scores for ranking video segment proposals of the localizer, and the localizer will produce word importance scores to help the captioner focus more on important and informative words.

Method	WS	M	C	R	B@1	B@2	B@3	B@4
DCE [14]	✗	4.82	17.29	-	17.95	7.69	3.86	2.20
DVC [16]	✗	6.93	12.61	-	12.22	5.72	2.27	0.73
Bi-SST [35]	✗	9.60	12.68	19.10	18.99	8.84	4.41	2.30
Masked-Transformer [56]	✗	9.56	-	-	-	-	4.76	2.23
SDVC [22]	✗	8.82	30.68	-	17.92	7.99	2.94	0.93
RUC AI-M ³ [30]	✗	11.28	14.03	-	16.59	9.65	5.32	2.91
Multi-modal MUTAN fusion [27]	✓	4.93	13.79	10.39	10.00	4.20	1.85	0.90
WS-DEC [10]	✓	6.30	18.77	12.55	12.41	5.50	2.62	1.27
EC-SL (Ours)	✓	7.49	21.21	13.02	13.36	5.96	2.78	1.33
EC-SL (less data)	✓	7.03	19.53	12.79	12.13	5.68	2.59	1.23

Table 3: Dense event captioning performances (%) of our proposed method and state-of-the-art methods on the ActivityNet Captions validation set. “WS” denotes “weakly supervised”.

4 EXPERIMENTAL RESULTS

4.1 Datasets

MSVD [2] and MSR-VTT [40] are two popular datasets for video captioning. MSVD contains 1,970 video clips that are split into 1200/670/100 clips for training, testing, and validation, respectively. Each clip has around 40 sentence annotations. MSR-VTT has 10,000 video clips that are split into 6,513/2,990/497 clips for training, testing, and validation, respectively. Each clip has 20 sentence annotations.

Charades-STA [11] and ActivityNet Captions [14] are commonly used to evaluate temporal sentence localization methods. Charades-STA has 6,670 videos and 16,125 localized sentence descriptions (multiple sentences per video). The training and testing set have 12,405 and 3,720 video-sentence pairs respectively. ActivityNet Captions has 19,994 videos and 100,000 localized sentence descriptions. The training, testing, and validation set have 10,009, 5,044, and 4,917 videos, respectively. ActivityNet Captions dataset is also used for evaluating dense event captioning methods.

4.2 Results

Effectiveness of Fine-Grained Information. For our proposed Motion Guided Spatial Attention, we demonstrate its effectiveness by ablation study on the MSVD dataset, and the results are shown in Table 1. As can be seen, compared to the basic spatial attention, our MGSA is significantly more powerful in terms of captioning performance. And the temporal relation established by the GARU is effective for both the basic spatial attention and our MGSA. To show the effectiveness of the Hierarchical Visual-Textual Graph, we also performed ablation study on its different layers and the results are in Table 2. As can be seen, each component in the HVTG is effective and they can be combined to achieve higher performances. Notice that the object-sentence subgraph is the most critical among these components.

Effectiveness of Modality Interaction. We demonstrate the effectiveness of our proposed modality interaction method by comparing it to existing methods that also focus on multimodal feature fusion. As shown in Table 4, when using the same set of features, our method (abbreviated as PMI-CAP) performs consistently better than existing methods. This can be mainly attributed to our design of modality interaction, which exploits more complementary information from multiple modalities.

Effectiveness of Bridging Captioner and Localizer. We aim to solve the dense event captioning task by bridging the captioner and localizer, and the experimental results are in Table 3. Compared

Method	Features	B@4	M	C
AF [12]	V+C	52.4	32.0	68.8
TDDF [51]	V+C	45.8	33.3	73.0
MA-LSTM [41]	G+C	52.3	33.6	70.4
MFATT [19]	R152+C	50.8	33.2	69.4
GRU-EVE [1]	IRV2+C	47.9	35.0	78.1
XGating [32]	IRV2+I3D	52.5	34.1	88.7
HOCA [13]	IRV2+I3D	52.9	35.5	86.1
PMI-CAP	V+C	49.74	33.59	77.11
PMI-CAP	G+C	51.55	34.64	74.51
PMI-CAP	R152+C	52.07	34.34	77.35
PMI-CAP	IRV2+C	54.68	36.40	95.17
PMI-CAP	IRV2+I3D	55.76	36.63	95.68

Table 4: Captioning performances of our proposed PMI and other state-of-the-art multimodal fusion methods on the MSVD dataset. Meaning of features can be found in [5].

to weakly-supervised methods, our proposed method has significant advantages for all the evaluation metrics, which is mainly due to the extra information passing pathways in our method. Note that there is still a gap between weakly-supervised and supervised methods, so the WS-DEC task remains very challenging.

5 CONCLUSION AND FUTURE DIRECTIONS

My research so far has found that exploiting fine-grained information in videos can boost the performance of high-level video understanding tasks, such as caption generation and sentence localization. These are tasks that bridge video and language and are more challenging, so a better understanding of the video contents is crucial. Besides, these two tasks can also be bridged. We have proposed a modality interaction method as their common foundation for better utilizing multimodal features in videos. To solve the weakly supervised dense event captioning task, we bridge the captioning and localization modules to form a cycle, and we outperformed SOTA methods by passing richer information between the two modules.

Exploiting fine-grained information is still worth exploring in the future, especially simultaneously building the spatial and temporal relations among the entities in a video. On the language side, how to utilize large text corpus or incorporate large-scale pre-trained language models are also interesting and challenging. Current video-and-language datasets are limited by their scales, and there exist sufficient unpaired video and language data, so it would also be interesting to study how self-supervised learning and transfer learning can be adopted to bridge video and language.

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